

Interview Q&A: deep learning

1. What is the difference between a convolutional neural network (CNN) and a recurrent neural network (RNN)?

A CNN is designed for image and signal processing tasks, using convolutional and pooling layers to extract spatial features. An RNN, on the other hand, is suited for sequential data such as text or time series, utilizing recurrent connections to capture temporal dependencies.

2. What is the purpose of batch normalization in deep neural networks?

Batch normalization is used to normalize the activations of each layer, reducing internal covariate shift and improving training stability.

3. What is the relationship between the number of layers and the capacity of a neural network?

Increasing the number of layers generally increases the capacity of a neural network, but can also lead to overfitting if not regularized properly.

4. What is the effect of using a non-saturating activation function, such as ReLU, on the training dynamics of a deep neural network?

Using a non-saturating activation function like ReLU can lead to faster convergence and better performance during training, but it also increases the risk of dying neurons.

5. What is the key difference between a generative adversarial network (GAN) and a variational autoencoder (VAE)?

A GAN consists of two neural networks that compete with each other, whereas a VAE uses a probabilistic approach to learn a continuous representation of the input data.

6. What is the relationship between the concept of 'mode collapse' and the training dynamics of generative adversarial networks?

Mode collapse occurs when a GAN fails to capture the full range of modes in the data distribution, resulting in limited diversity in generated samples.

7. How does the concept of 'mode collapse' relate to the training dynamics of generative adversarial networks?

Mode collapse occurs when a GAN fails to capture multiple modes or patterns in the data distribution, resulting in limited diversity and variability in the generated samples.

8. What is the effect of using a large learning rate on the convergence properties of stochastic gradient descent?

A large learning rate can lead to oscillations or divergence, as it may overshoot the optimal solution.

9. What is the relationship between the concept of 'exploding gradients' and the use of gradient clipping in deep neural networks?

Gradient clipping is a technique used to prevent exploding gradients by limiting the maximum allowed value of the gradient norm, thus preventing large updates that can cause instability in training.

10. How does the use of weight normalization affect the optimization dynamics of deep neural networks?

Weight normalization can improve optimization by reducing the effect of large weights on the gradient, but it also introduces additional computation and may not be necessary for all models.

11. How does the use of attention mechanisms in deep learning models affect the computational complexity of sequence-to-sequence tasks?

Attention mechanisms can significantly reduce the computational complexity by only considering relevant parts of the input sequence, rather than processing the entire sequence at once.